Unit 4 · Outcome 2 · Energy Sources

Sustainability principles, applied across the energy supply chain

KK07 VCE Environmental Science · Units 3 & 4 · Exam study summary

> Every other dot point in this chapter has handed you a piece of an energy source — what it is (KK01–02), what burning it does to the carbon cycle (KK03), how fast we are using it (KK04), how efficiently it converts (KK05), and what the laws of thermodynamics permit (KK06). KK07 is where you assemble those pieces into a judgement. It asks you to run the six sustainability principles like a checklist across the five stages an energy resource passes through on its way from the ground to your power point.

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.

THE SUPPLY CHAIN

01 Five stages, one resource

The study design names the stages precisely: accessing → extracting → processing → transporting → using. Learn them as a sequence, because exam questions love to pin a principle to one specific stage ("evaluate the user-pays principle at the extraction stage").

The scene below follows a single energy resource down the Tarra Cove corridor in Gippsland — the fictional coastal town we have used across this chapter, sitting beside the real Latrobe Valley brown-coal fields and the Bass Strait gas province. Click a stage to fly the camera to it and read what each stage physically does before we judge it.

> • **NOTE**
>
> 3D view unavailable in this browser — the stage cards and readout below work exactly the same.

THE YARDSTICK

02 The six sustainability principles

These are the same six VCAA principles you met in U3·O2 — but here you swap "environmental management" for "energy". Memorise the one-line "energy test" on each card: that is the lens you point at every stage.

Two principles students confuse every year. Intergenerational equity is fairness between generations — today's energy use must not leave a degraded planet or an un-rehabilitated mine for people not yet born. Intragenerational equity is fairness within today's generation — energy access, costs and harms shared fairly between rich and poor, city and region, here and overseas. Same root word (equity = fairness); different time direction. Inter = between (generations); intra = within (one generation).

THE WHOLE PICTURE

03 Principle × stage: the interactive matrix

This grid is the dot point. Six principles down the side, five stages across the top. Click any cell to see how that principle applies at that stage, with an Australian example. This is the structure to reach for whenever a question says "apply the sustainability principles to…".

REAL AUSTRALIAN CASES

04 Examples that earn marks

Examiners reward specific, current, Australian examples over vague generalities. Lock these three — one each for the principles students find hardest to illustrate.

Victoria permanently banned fracking and coal-seam-gas exploration and wrote the ban into the state constitution (2021). Even though the science on groundwater risk was contested, the government acted to prevent serious, potentially irreversible harm to farmland and aquifers. That is the precautionary principle at the accessing/extracting stage.

Australia's Safeguard Mechanism caps the emissions of facilities over 100,000 t CO_2-e/yr; baselines fall ~4.9% a year and emitters over the line must buy carbon credits (~$37/t in 2026). Polluters pay for the climate cost of processing & using fossil energy — instead of dumping it on the public.

The Hazelwood mine (closed 2017) is being rehabilitated into a full pit lake — a "safe, stable, sustainable, non-polluting" landform — so a 135 m-deep void with fire and collapse risk is not inherited by future generations. Rehabilitation bonds and a 2026 trailing-liabilities law force the company, not future taxpayers, to pay.

The user-pays / intergenerational tension to remember: Engie's Hazelwood rehabilitation bond is about $289 million, but the estimated full cost is roughly $743 million. When the bond is smaller than the real bill, the shortfall risks falling on the public and the future — a live example of a principle being only partly upheld. "Partly upheld, because…" is exactly the nuance that lifts an evaluate answer.

COMPARE TWO PATHWAYS

05 Brown coal vs solar + storage

"Compare" and "evaluate" questions want you to weigh one source against another across the principles. Toggle the two pathways and watch the pressure each one puts on each principle. Taller bar = more strain on that principle (worse for sustainability).

> ● **NOTE**
>
> 3D view unavailable — the principle bars below update the same way.

ACTIVE RECALL

06 Which principle is being upheld?

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

READ THE QUESTION

07 Command-word decoder

> ◈ **COMMAND WORDS**
>
> The verb tells you what shape your answer must take. Mis-reading "evaluate" as "describe" is the single most common way to lose marks on this dot point.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

DON'T FALL FOR THESE

08 Six exam traps

⚠ **EXAM TRAPS**

Each of these has cost real students real marks. Open one, then close it and try to recall the fix.

P · E · L

Build a body paragraph that scores

High-band answers on this dot point follow P oint → E vidence → L ink. Make a point (name the principle & stage), give evidence (a specific Australian example with a figure), then link back to "upholding" or "undermining" the principle. Type below — the meter lights up as you hit each move.

± WORKED EXAMPLE

Same question, two answers

Q. "Apply one sustainability principle to the extraction stage of an energy resource of your choice." (3 marks) — watch how the weak answer describes, while the strong answer applies and links.

Coal mining is bad for the environment because it digs up the land and makes pollution. We should use the precautionary principle and the user-pays principle because polluters should pay. Sustainability is important for the future so we should look after the environment for the next generation.

● **NOTE**

Why it loses marks: names principles but never applies them to the extraction stage; no Australian example; "bad for the environment" is not ecological integrity; lists three principles vaguely instead of developing one. Mark: identifies a principle (1), but no application and no evidence.

The user-pays principle requires those who extract a resource to bear its full environmental cost. At the extraction stage of Victorian brown coal, the operator of a Latrobe Valley mine must lodge a rehabilitation bond with the State so that the cost of returning the 135 m-deep void to a safe, stable landform is paid by the miner, not the public. This upholds user-pays by internalising the rehabilitation cost into the price of the coal — though it is only partly upheld where the bond ($289M) is below the estimated full cost ($743M).

● **NOTE**

Why it scores: defines the principle (1), applies it to the named extraction stage with a specific Australian mechanism + figure (1), and links explicitly to "uphold", with a nuance that shows evaluation (1).

★ PRACTICE EXAM

Five VCAA-style questions · 20 marks

★ **PRACTICE & SELF-MARK**

Attempt each in your logbook first, then reveal the model answer and marking scheme. Award yourself marks honestly — the dock at the bottom tallies your total and saves it on this device.

≣ ONE-SCREEN SUMMARY

KK07 cheat sheet

- Accessing — finding it & getting legal/physical entry
- Extracting — mining, drilling, or harvesting the flux
- Processing — refining, drying, enriching, converting
- Transporting — pipelines, transmission, ships, trucks
- Using — final conversion to heat / motion / electricity

- Ecological integrity — protect ecosystems & their function
- Resource-use efficiency — most useful energy per unit input
- Intergenerational equity — fair to future generations
- Intragenerational equity — fair within today's generation
- Precautionary — act on serious/irreversible risk under uncertainty
- User-pays — those who use/pollute bear the full cost

- Precautionary → Vic fracking/CSG ban in the constitution
- User-pays → Safeguard Mechanism (>100kt) & rehab bonds
- Intergenerational → Hazelwood pit-lake rehabilitation
- Efficiency → cutting transmission & conversion losses (KK05/06)
- Intragenerational → just transition for Latrobe Valley workers

- Point: name the principle + the stage
- Evidence: a specific Australian example with a figure
- Link: state whether it upholds or undermines — and how fully
- Evaluate? add the "partly, because…" tension
- inter =between gens · intra =within one gen

ECOLOGICAL INTEGRITY RESOURCE-USE EFFICIENCY INTERGENERATIONAL EQUITY INTRAGENERATIONAL EQUITY PRECAUTIONARY USER-PAYS

Ecological integrity Resource-use efficiency Intergenerational equity Intragenerational equity Precautionary User-pays

Every other dot point in this chapter has handed you a piece of an energy source — what it is (KK01–02), what burning it does to the carbon cycle (KK03), how fast we are using it (KK04), how efficiently it converts (KK05), and what the laws of thermodynamics permit (KK06). KK07 is where you assemble those pieces into a judgement. It asks you to run the six sustainability principles like a checklist across the five stages an energy resource passes through on its way from the ground to your power point.

Think of it as a quality-control inspection of an energy supply chain. At every stage — the mine, the well, the refinery, the pipeline, the wall socket — you ask the same six questions. The answers are what separate a coal pathway from a solar one, and they are exactly what examiners want you to apply, compare and evaluate.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Coal mining is bad for the environment because it digs up the land and makes pollution. We should use the precautionary principle and the user-pays principle because polluters should pay. Sustainability is important for the future so we should look after the environment for the next generation. Why it loses marks: names principles but never applies them to the extraction stage; no Australian example; "bad for the environment" is not ecological integrity; lists three principles vaguely instead of developing one. Mark: identifies a principle (1), but no application and no evidence.

✓ STRONG / MODEL ANSWER

The user-pays principle requires those who extract a resource to bear its full environmental cost. At the extraction stage of Victorian brown coal, the operator of a Latrobe Valley mine must lodge a rehabilitation bond with the State so that the cost of returning the 135 m-deep void to a safe, stable landform is paid by the miner, not the public. This upholds user-pays by internalising the rehabilitation cost into the price of the coal — though it is only partly upheld where the bond ($289M) is below the estimated full cost ($743M). Why it scores: defines the principle (1), applies it to the named extraction stage with a specific Australian mechanism + figure (1), and lin

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.